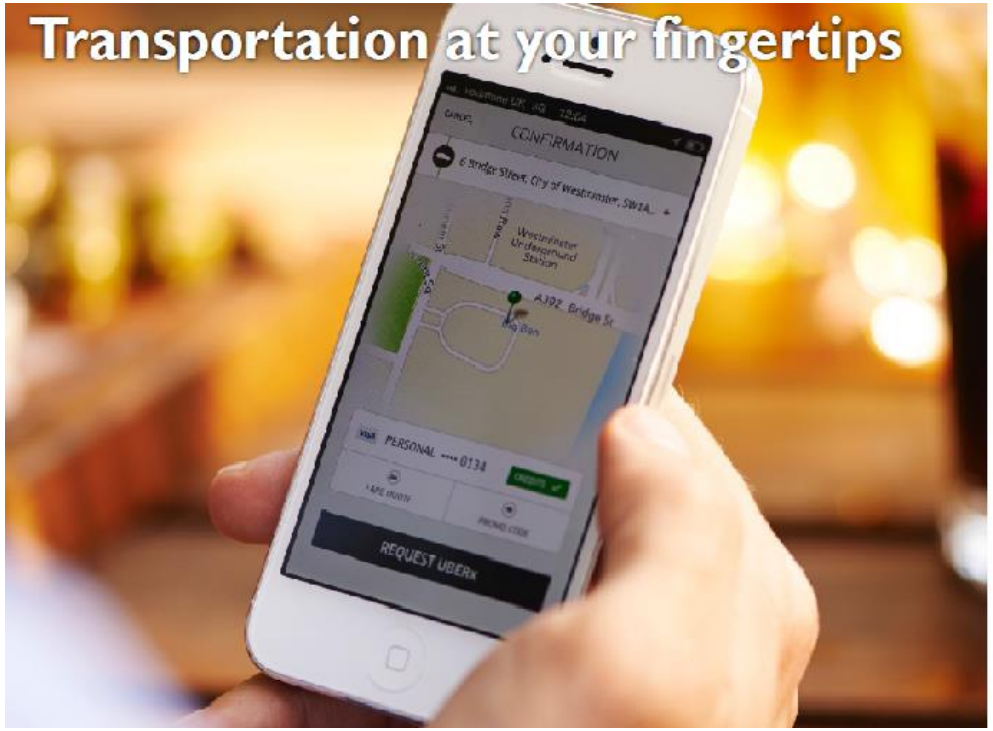
UBER.com – A case study



Where lifestyle meets logistics….

- Prajakta Soman

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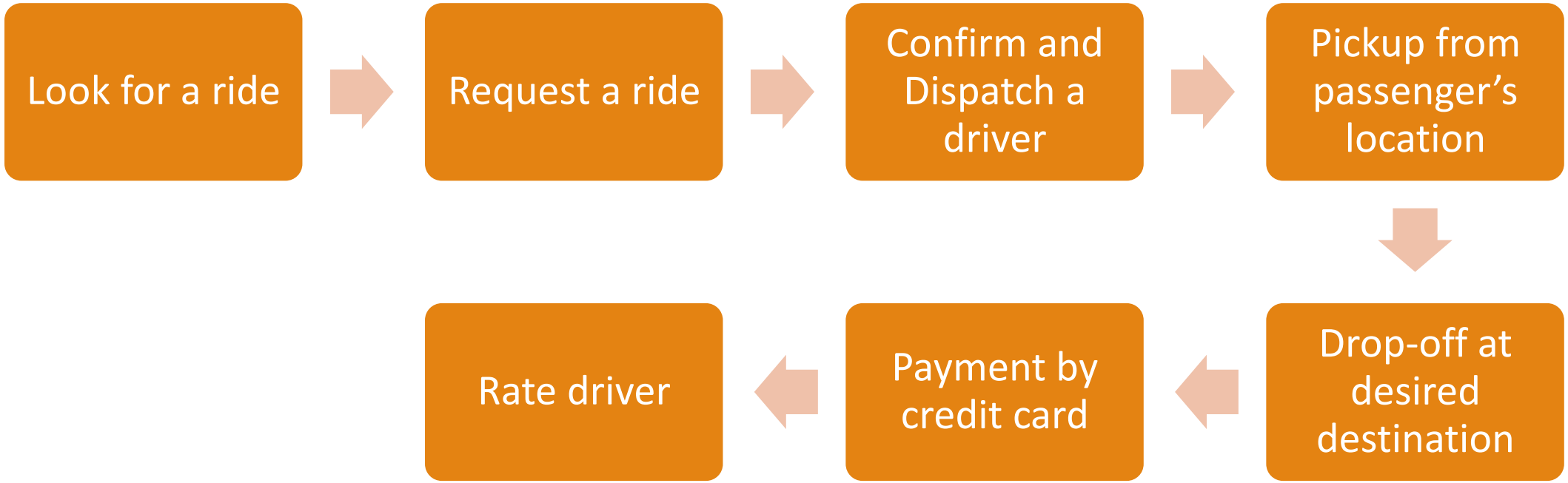
About Uber Technologies, Inc

- American multinational online company (NYSE: UBER)
- Headquartered in San Francisco, California
- Provides mobility as a service (ride-hailing, food delivery, package delivery, courier)
- Develops, markets and operates Uber mobile app
- Provides on- demand service to the user
- Founded in 2009
- Spans over 72 countries and 10K + cities in the world
- 100,000 drivers associated with Uber

Process flow



Business Model

- Uber doesn’t employ any taxi drivers nor owns any taxi
- It connects the passenger (demand) and taxi driver (supply) and takes a percentage of the fee from the fare
- Taxi driver –
  - Anyone with a driving license and a car can apply for an Uber driver
  - Enlisted drivers are given an Uber iPhone. Drivers are Uber Partners
- Passenger –
  - Registered Uber users download the Uber app to their phones
  - They call a taxi via the Uber app
  - Can also track the taxi on their phone as it approaches
- Fare and Payment –
  - Fare set by Uber, based on car type, distance and peak hour
  - Payment by Credit Card
  - 80% share to the driver, 20% to Uber

Success Story and Motivation

- Uber's business is rooted firmly in Big Data
- Business Model is based on Big Data principle of crowd sourcing
- Big Data informed **Surge Pricing** – uses predictive modelling to estimate demand in real time
- Uses Data Science and Artificial Intelligence to optimize its operations and predict where its riders want to go.
- Uber Pool Service by finding look-alike trips using Big Data
- Use of smart phone applications, GPS, Google maps and social networking
- Uber shares detailed trip level data, without compromising on the privacy of riders and drivers.
- Municipal partners use this data in Smart City projects like traffic planning, congestion reduction, creation or reduction of parking, impact of events and disasters on city transportation.

Surge Pricing

- Dynamic pricing/ surge pricing is a novel concept that has been popularized by Uber and being adopted in other verticals as well, such as food delivery.
- Whenever there is a higher demand for cabs than what can be served at that moment (for example, at the airport after a flight lands), the fare goes up based on a **surge price calculation algorithm.**
- This algorithm adjusts rates based on a number of variables, such as time and distance of your route, traffic and the current rider-to-driver demand.
- Some drivers move to the surge region to earn extra (increases supply) and some passengers opt to wait to get a ride (reduces demand).
- This way Uber is able to manage the demand-supply mismatch situation better.
- Drivers make more money, Uber makes more money and customers spend more money (but get urgent rides).

Data @ Uber

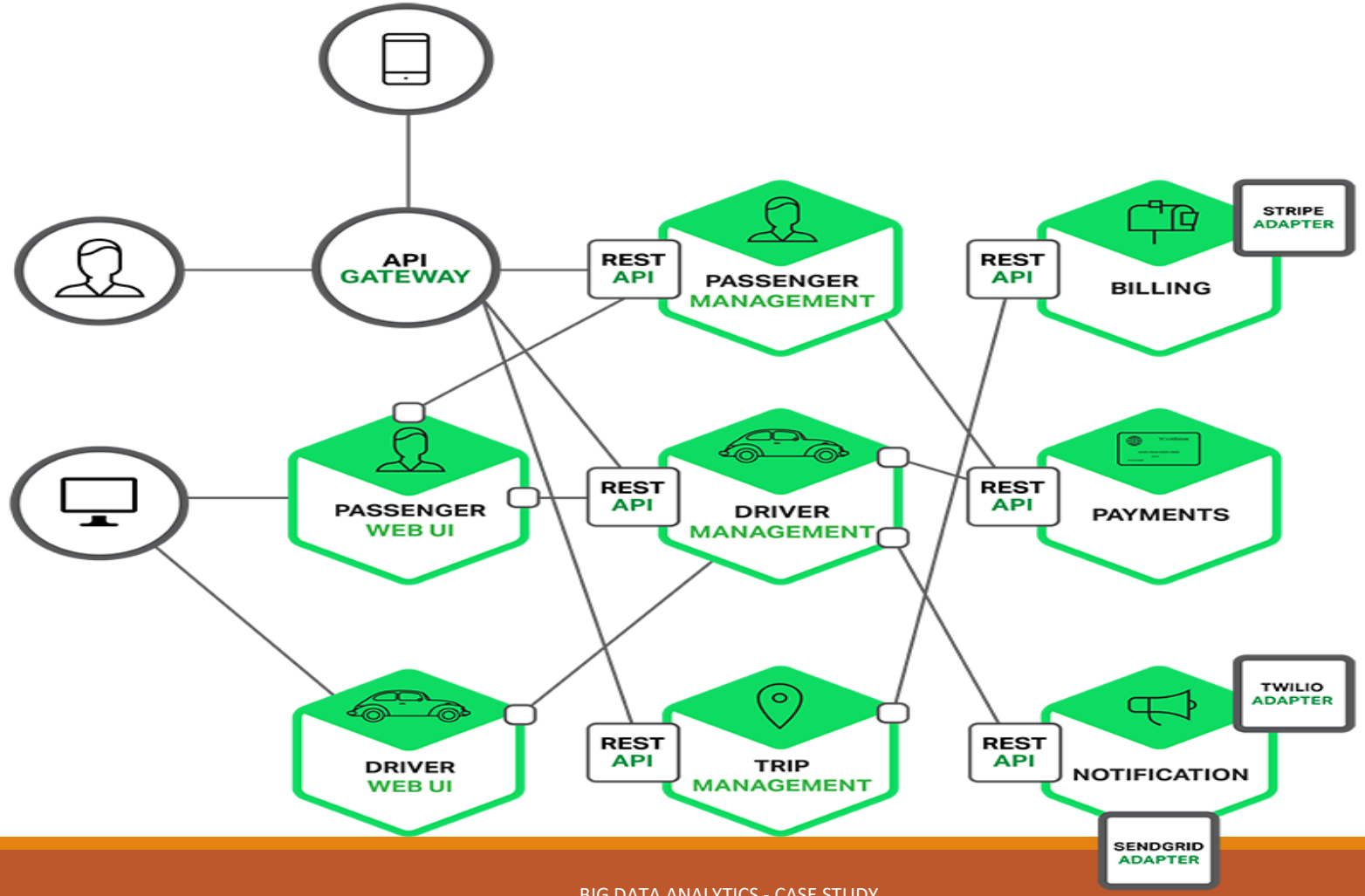
- Multi-dimensional data
- Over 10,000 hexagons in the city
- 300 cities
- 7 vehicle types
- 1440 minutes in a day (24 hours service available)
- 13 driver action-states
- 1 day of data =300 x 10,000 x 7 x 1440 x 13 = 393 billion possible combinations!
- Application emits events
- Billions of events per day, at least dozens of fields in each event =
- Uber has generated over 100s of PBs of data

Big Data

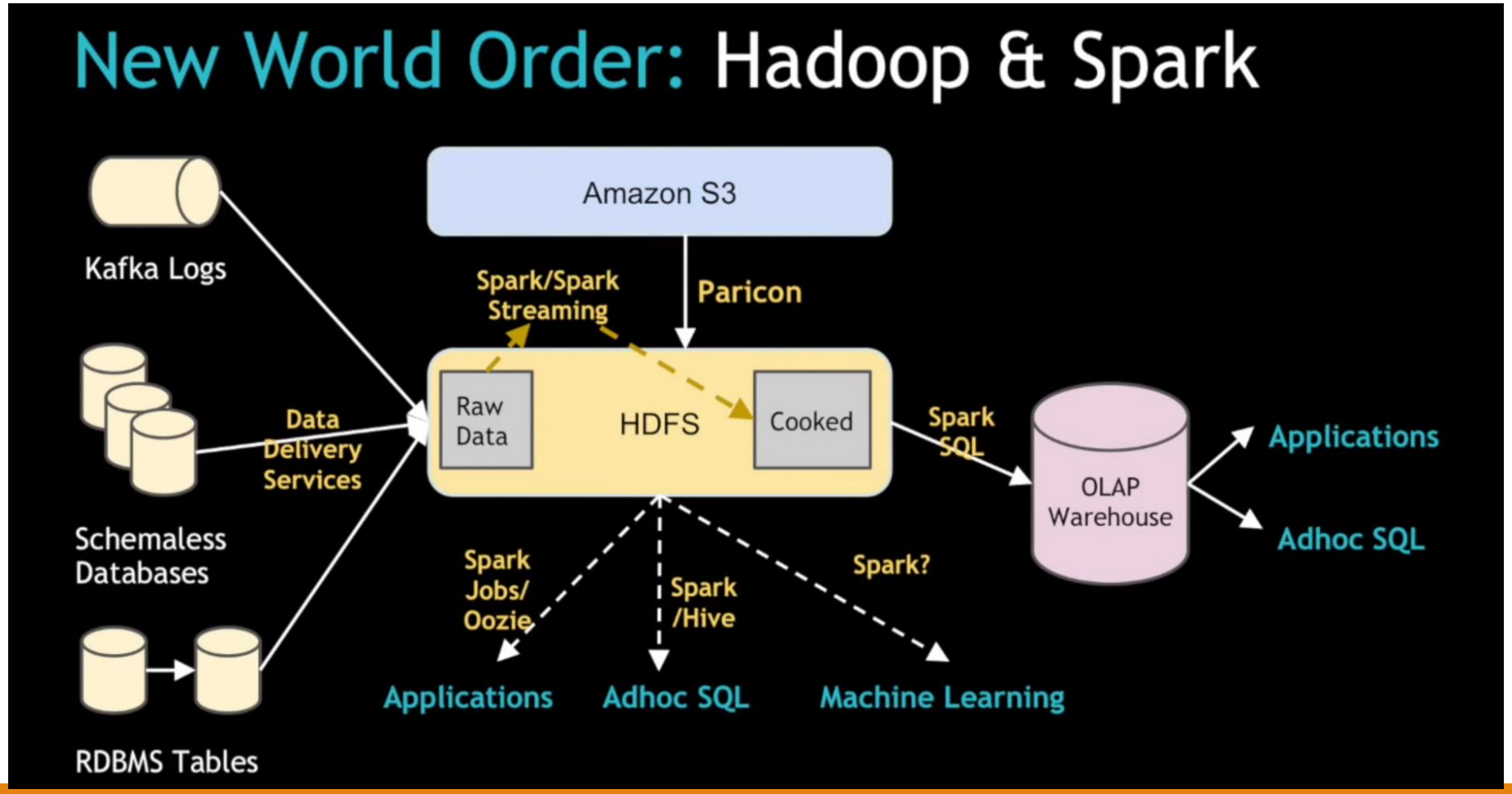
Data Attributes

Trip UUID	• Universal Unique Identifier with 128 bit value. e.g. **de305d54-75b4-431b-adb2-eb6b9e546014**
Driver state	• Open, off duty, accepted, rejected, dispatched, dispatch pending, client cancelled, driver cancelled
Vehicle type	• UberX, UberXL, UberPOOL, UberBlack
Timestamp	• Current date and time
Latitude	• pickup and drop
Longitude	• pickup and drop
Fare	• Base fare + time + distance

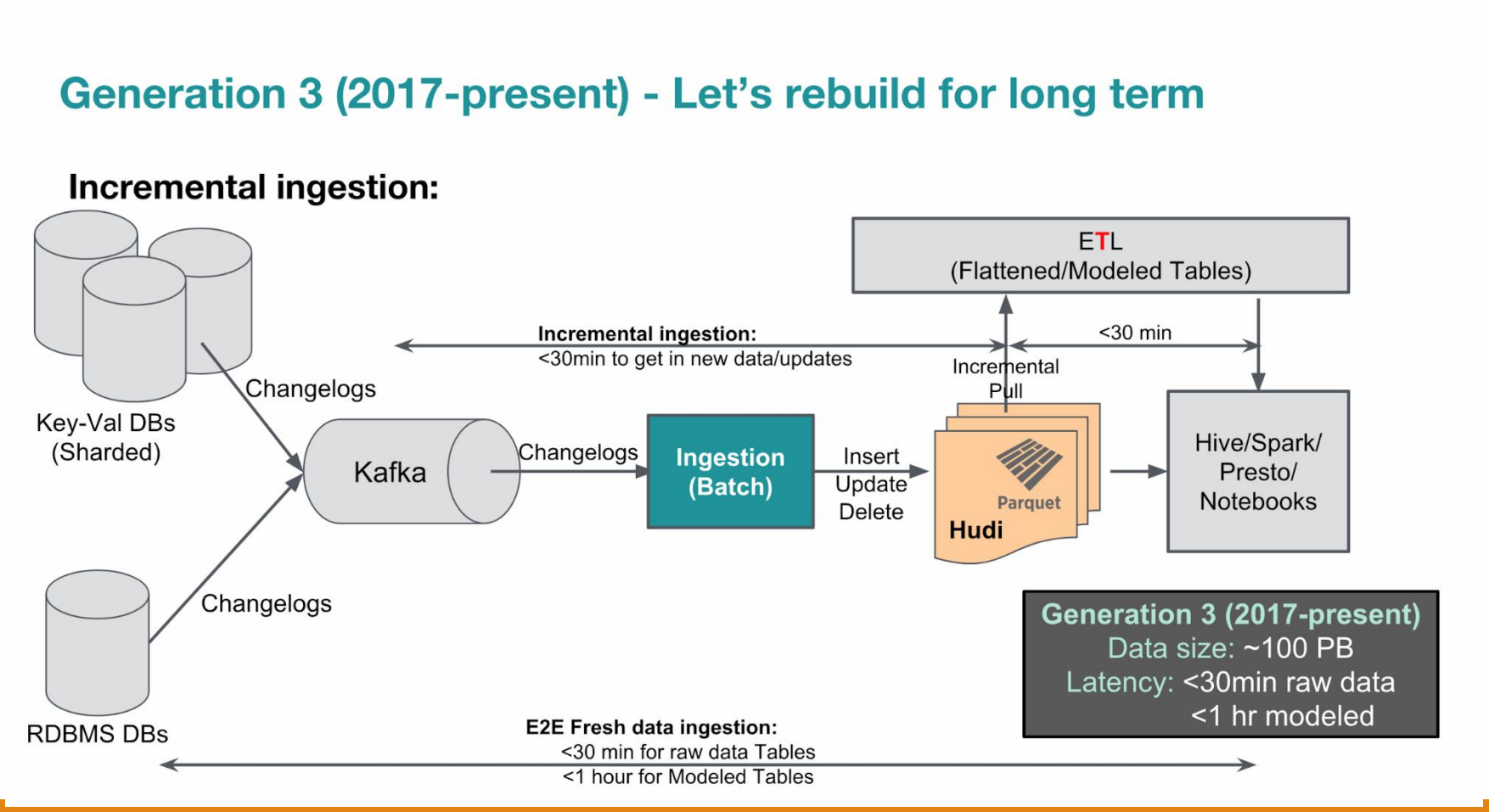
Microservices Architecture Pattern



OLAP Architecture @ Uber



Current framework at Uber

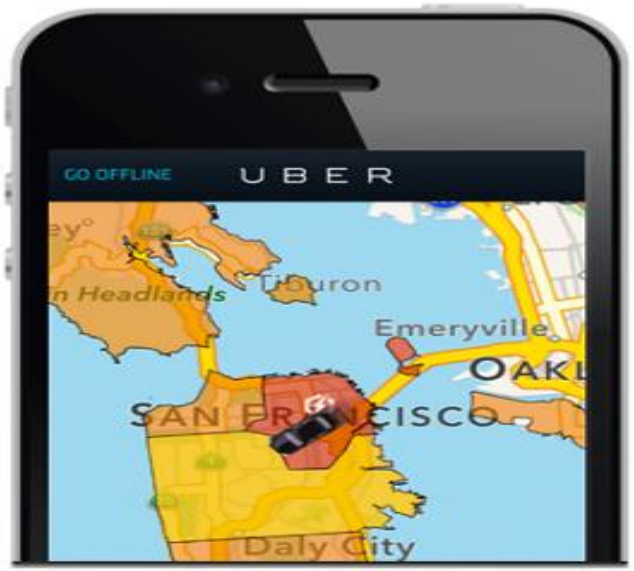


OLAP @ Uber

- Heatmap, Geo-Temporal Analytics
- Top N, Count, Avg, Sum, Percent
- Adhoc Queries:
  - Which riders request a pickup from 100 miles apart within a half hour window?
  - How many drivers cancel a request 10 times in a row within a 5-minute window?
  - Any combination of dimensions….
- Histogram, Supply Vs Demand
- City Management, Warn drivers about possible accidents
- Drivers Payout

Heatmap

Make sure you use these heatmaps to position yourself in the busy areas to maximise your earnings!



– MAP COLOURS SHOW CURRENT DEMAND –

NO COLOUR = NORMAL DEMAND

YELLOW = MODERATE DEMAND

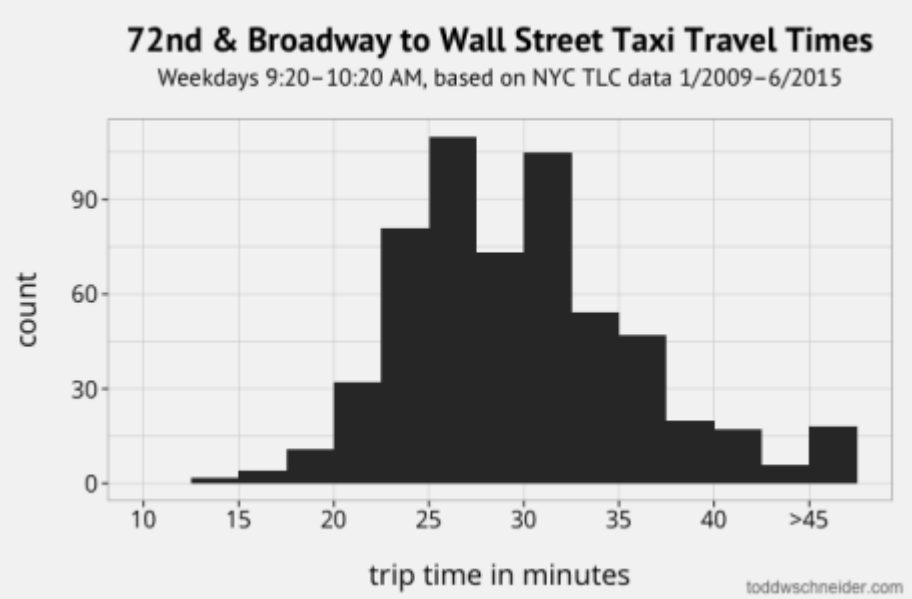
ORANGE = HIGH DEMAND

RED = SURGING

The pulse of a city….

Histogram – Travel time

- Source A
- Destination B
- Timing:
  - 9:20 to 10-20 a.m. weekdays
- Period:
  - 1/2009 – 6/2015
- Observation:
  - Mean Travel Time: 29.8 min
  - Median: 29 min
- Conclusion:
  - 50% of the trips were made within allotted time of 30 minutes



Uber Dataset – .csv file, 300 MB, year 2014, 4451635 rows

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uber-trip-nyc-14.csv

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Glossary

Kafka - Kafka is a popular open source system for managing real-time streams of data from websites, applications and sensors that is now being used as fundamental infrastructure by thousands of companies. Kafka-based data pipelines are used to feed Hadoop clusters and data warehouses

Geofencing - A geofence is a virtual barrier. Programs that incorporate geo-fencing allow an administrator to set up triggers so when a device enters (or exits) the boundaries defined by the administrator, a text message or email alert is sent. Uber uses this to find out where lots of people are requesting rides at the same time. For this latitude-longitude is retrieved from customers mobile phone and then it is determined which geofence the customer belongs to. Also used for defining areas with specific requirements such as airports.

Crowdsourcing - Business term coined in 2005 - the process of obtaining needed services, ideas, or content by soliciting contributions from a large group of people, and especially from an online community, rather than from traditional employees or suppliers.

Glossary (continued…)

Microservices Architecture Pattern - This design pattern enforces the development of small services dedicated to specific, well-encapsulated domain areas. Each service can be written in its own language or framework, and can have its own database and can be independently deployed.

REST APIs – REST (Representational State Transfer) is a decoupled Architecture Style enabling lighter weight communication between producer and consumer. Used in mobile applications, social networking, cloud systems.

Spark – Open source Big Data Processing Framework for managing diverse datasets, in batch as well as real-time stream. Runs on HDFS.

Hive - Apache Hive is a data warehouse infrastructure built on top of Hadoop for providing data summarization, query, and analysis (OLAP tool)

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